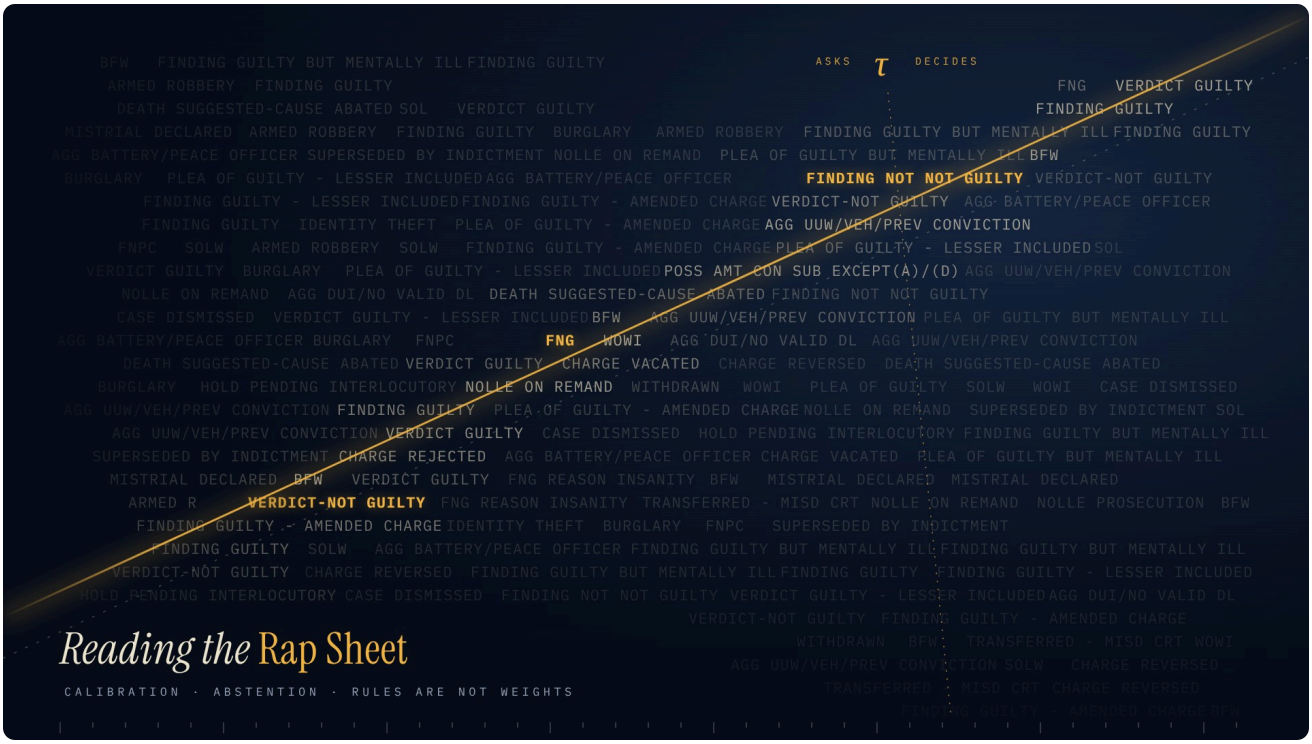
Reading the Rap Sheet

Teaching a small model to read charge text — and knowing when it shouldn't decide

A small model reads the charge. A versioned lookup reads the outcome. A threshold decides who gets to decide.



> *Finding Not Not Guilty.*
> — a real disposition string, 2,437 times

Steven B Miers · October 21, 2026

Three strings in one public court table mean the same thing: `FNG`, `Verdict-Not Guilty`, and, 2,437 times, `Finding Not Not Guilty`. Read literally, *not not guilty* means **guilty**. Nobody meant that. It's what **noisy data** looks like up close: one fact recorded three ways, one of which says the opposite of what it means. A human decodes it in half a second; a regex never will. Above all three sits `Nolle Prosecution` at 714,681 rows, two-thirds of the table. Thirty-six outcome strings in all, and three appear exactly once.

Here's what I built. A court record has two text fields that matter: the **charge** (what someone was accused of) and the **disposition** (what the court did about it). A small model reads the charge, sorts it into one of the prosecutor's offense categories, and says *how sure* it is, and means it. Below a certain confidence it doesn't decide; a person does. The disposition gets no model at all. It gets a lookup table a human wrote, versioned, and can defend.

Two fields, two problems, and the mistake most AI-for-records products make is solving both with the same hammer. Thirty-six strings is a *lookup*. Thirteen hundred charge titles, with new ones every year, is a *model*. The transformer ties the statute book on accuracy, beats it on the rare categories once I stopped starving them, and earns the right to decide alone on three categories out of 68. That last part is the interesting one.

Every number below traces to a run manifest. This is a reference implementation of a methodology, not a screening tool, and nothing in it tries to work out who a record belongs to. It is the worked example for Part 3 of my series *Semi and Fully Autonomous AI Systems*: a model earning autonomy one decision class at a time.

🧠 Part 1: Concepts

🔤 Two vocabularies, one record

The disposition is a closed vocabulary, and a noisy one. Thirty-six strings for five outcomes: convicted, acquitted, not prosecuted, vacated, not final. Some differences are real procedure (a dismissal isn't a nolle); plenty are noise, the same outcome spelled by different clerks in different decades. A person can read them all in ten minutes, decide each one once, and write the decisions down with a version number. `Finding Not Not Guilty` maps to *acquitted*, with a note saying why. When someone disputes a line on their record, the answer is a rule with a version on it, not a logit. A model would have to *learn* that a double negative is a typo, and could quietly unlearn it at the next retrain.

The charge title is an open vocabulary, mostly. There are 1,365 distinct titles, abbreviated by clerks over decades (`RET THEFT/DISP MERCH/<$300/PRECONV`, where `<$300` is the part that matters). You *could* hand-write that lookup, and it would be stale by spring: new statutes add titles, and the legacy case system spelled the old ones differently. A lookup memorizes. A model has a chance at the title it has never seen and, if it's calibrated, the decency to admit when it isn't sure.

The label column isn't innocent either. One of its 78 values, `PROMIS Conversion` (nearly ten thousand rows), isn't an offense; it marks records migrated from the old case system. Same disease: noise that looks like data.

Rules are not weights. The closed vocabulary goes in a lookup a human owns. The open one goes in a model a human supervises.

🌡️ What calibration buys you

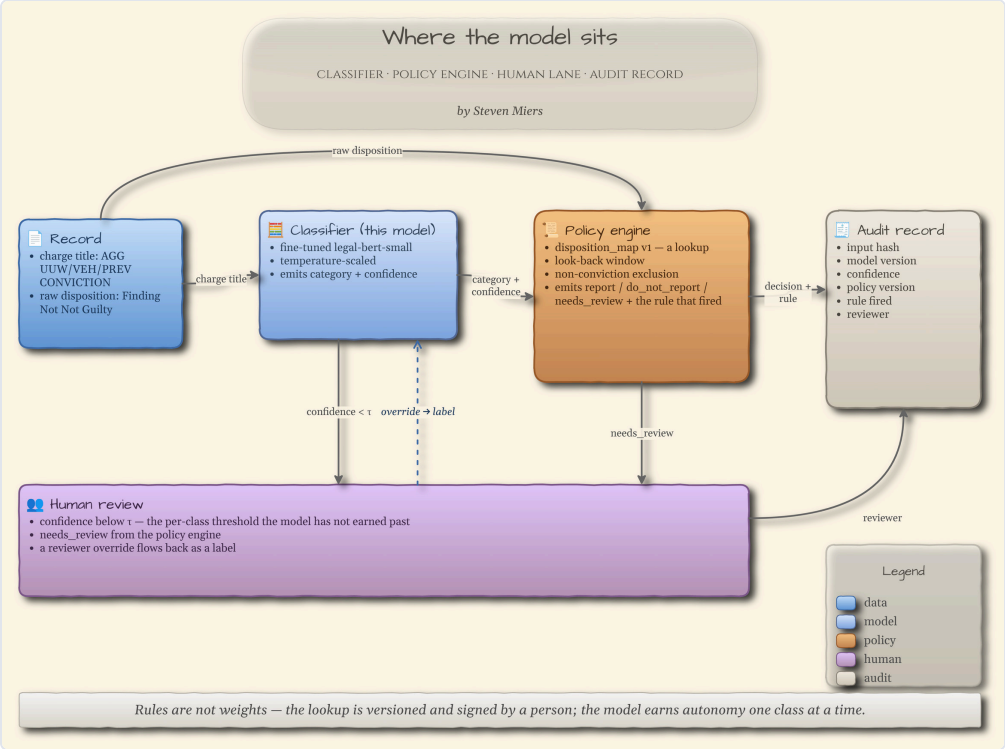
A softmax output is a score, not a probability, until you make it one. Fine-tuned transformers are often overconfident: the "92%" bucket is right 80% of the time, like a weather app that has never once doubted itself. **Temperature scaling** (Guo et al., 2017) fits one number T on held-out data

and divides the logits by it. The top answer never changes; only the confidence gets honest, and honest confidence is what the next step spends as a *budget*.

🚦 What abstention buys you

With honest confidence you can set a threshold τ: *below τ, a person decides.* That's **selective classification** (Chow, 1970; Geifman & El-Yaniv, 2017), reported as a risk–coverage curve: for each share of the traffic the model handles alone, what error rate are you accepting? This piece uses a 0.5% budget, applied **per class**. A category earns its threshold only if its validation error stays inside the budget with 95% confidence (the upper end of a Wilson interval), because zero mistakes in thirty rows doesn't prove a half-percent error rate; it proves you looked at thirty rows. What comes out is a table of the categories the model has *earned* the right to decide alone.

🏛️ Where the model sits



D1: where the model sits

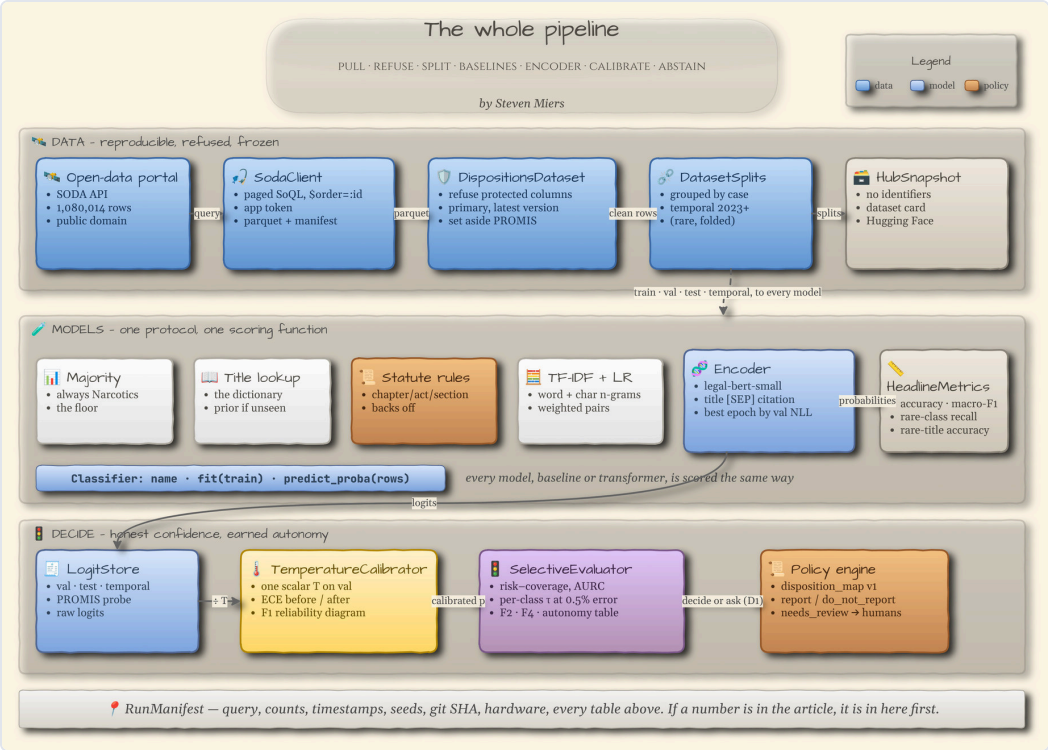
Three slots and a lane. The **classifier** emits a category and a calibrated confidence. The **policy engine** reads the disposition through the versioned map, applies the look-back window and the non-conviction exclusion, and emits `report`, `do_not_report` or `needs_review`, *with the rule that fired*. The **human lane** gets everything below τ and everything the policy flagged, and a reviewer's override comes back as a training label. The **audit record** keeps enough to replay any decision.

🗃️ The data, in three sentences

One county's public-domain prosecutions table: about 1.08 million charge-level rows, de-identified by the publisher and pulled by query from a SODA open-data portal. The publisher stopped maintaining it in December 2024, but the portal's row timestamps moved as late as April 2026, so every number here traces to a run manifest (query, counts, source timestamp, seeds, code version). `race`, `gender`, `age_at_incident`, `judge`, `incident_city`, `law_enforcement_agency` and `unit` are never pulled, nor is `offense_category` (the pre-update label, which would leak the answer), and the loader refuses any frame that carries one.

🏗️ Part 2: Build

🗺️ The whole pipeline



D2: the whole pipeline

Three lanes, one protocol, and a receipt for everything:

Component	Job	Why it's here
🎣 **SodaClient**	portal → parquet + manifest	A query is a recipe anyone can rerun; a CSV is a snapshot of somebody's afternoon

Component	Job	Why it's here
🛡️ **DispositionsDataset**	refuse, clean, set PROMIS aside	"We never request them" is a promise; the error is a check
🔗 **DatasetSplits**	train / val / test / 2023+	No case straddles the test line, and 2023+ is "the world moved"
📊 **Majority**	always *Narcotics*	The floor: 37%, and embarrassingly hard to beat on a bad day
📖 **Title lookup**	the title's usual label	The dictionary. Can't beat it? You're remembering, not reading
📜 **Statute rules**	chapter, act, section → label	The rule a lawyer could write down
🧮 **TF-IDF + LR**	word and character n-grams	A long record of embarrassing transformers on short, stylized text
🧬 **Encoder**	`title [SEP] 720 ILCS 5/24-1.6` → 68 logits	`legal-bert-small`: a 6-layer BERT, 35M parameters, one new 68-way head, every weight fine-tuned
🌡️ **TemperatureCalibrator**	validation logits → T	Makes "92%" mean 92%
🚦 **SelectiveEvaluator**	probabilities → τ per class	Decides what the model may decide alone
📍 **RunManifest**	everything → JSON	If a number is in this document, it was in the manifest first

Every model implements the same three methods (`name`, `fit`, `predict_proba`) and is scored by the same function. Nothing gets graded on a curve.

🎣 Pull by query, not by download

The pull is a paged query with a stable sort, so page 14 means the same rows every time, and the protected columns never appear in the select list: it's hard to leak what you never downloaded. Rows reported and rows received both say 1,080,014.

🔗 Split by case, then by time

Charges in one case share titles, dates and wording. Let them straddle train and test and the model gets graded on homework it has already seen, so the split is grouped by case and stratified by label. Everything received from 2023 on is held out entirely.

📜 The policy is a file

```
acquitted: [FNG, Verdict-Not Guilty, Finding Not Not Guilty, FNG Reason Insanity]
```

Thirty-six strings, five outcomes, a version number, and a loader that reports what it doesn't recognize instead of guessing. A lookup that guesses is a model with worse documentation.

🧩 The transformer

The transformer is the reader, not the decider. `legal-bert-small` (Chalkidis et al., 2020) is a six-layer BERT encoder with a 512-wide hidden state, 35M parameters, pre-trained by masked-language modelling on legislation, contracts and case law. It was never asked to classify anything. Loading it as a sequence classifier keeps every encoder weight, throws away the masked-language-model head, and bolts on one new linear layer from the `[CLS]` vector to 68 logits: 512 × 68 weights and 68 zero-initialized biases, about 35k parameters that start from nothing on top of 35M that don't. The head is the only new piece. No new vocabulary, no adapters, no frozen layers; the whole network trains end to end, and the head's bias quietly learns roughly the log of how often each class shows up.

In: a sentence pair. Segment A is the title exactly as the county spells it, `AGG UUW/VEH/PREV CONVICTION`; segment B is the citation, `720 ILCS 5/24-1.6`. The tokenizer lowercases and keeps punctuation, which matters when `<$300` and `>$300` are different charges. Nothing else goes in: no disposition, no date, and never a protected column.

Out: 68 logits in a fixed label order, 67 named categories plus one fold bucket for anything under 20 training rows. Softmax turns them into a probability per category; the top one is the prediction, and its probability is the confidence Part 3 spends. Everything downstream (the logit adjustment, the temperature, the thresholds) lives in a file next to the weights, not inside them: a working classifier with none of the policy, or all of it, by choice.

🧬 The training loop

The 266,439 training rows hold only 1,726 distinct (title, citation) inputs, so the loop encodes each one once and draws them in proportion to how often they occur. Same distribution as drawing rows, at coffee-break prices. The epoch with the lowest validation *negative log-likelihood* wins, because calibration rewards honest probabilities over confident ones. Three epochs took about ten minutes on an Apple-silicon GPU, and a rerun reproduced every number to the last decimal. (Hold on to "in proportion to how often they occur." It comes back to bite.)

📊 Part 3: Evaluation

📏 The headline table

Four baselines and two runs of the encoder: the first run, and the rerun after I found out what the first one was doing wrong. Two test sets: the stratified test split (33,305 rows) and everything received from 2023 on (19,973). *Rare-class recall* averages the 35 categories with fewer than 500 training rows; *rare-title accuracy* covers titles seen fewer than five times in training.

Model	Test acc	2023+ acc	Macro-F1	Rare-class recall	Rare-title acc
📊 Majority (floor)	37.5%	31.0%	0.008	0.0%	5.8%
📖 Title lookup	95.2%	94.3%	0.826	74.8%	59.5%
📜 Statute rules	**96.4%**	**96.0%**	0.861	78.0%	63.7%
🧮 TF-IDF + LR	95.3%	94.3%	0.823	75.2%	**69.0%**
🧬 legal-bert-small, first run	96.1%	95.5%	0.688	49.4%	60.5%
🧬 legal-bert-small, class-tempered	**96.4%**	95.8%	**0.880**	**85.2%**	64.7%

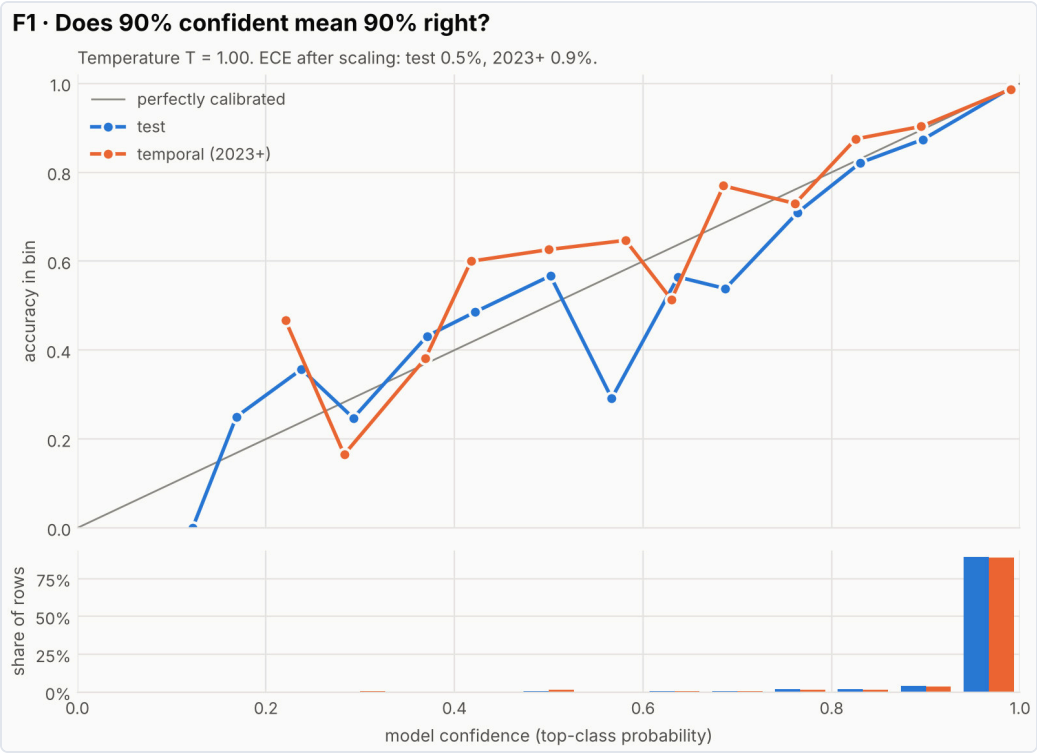
On accuracy, the statute book and the transformer tie: 96.4% each, seven test rows apart. Everywhere the small categories count, the transformer wins: macro-F1 0.88 against 0.86, rare-class recall 85% against 78%. Getting there took a rerun.

Everyone is fighting over the last point and a half. Even a perfect memorizer of every (title, citation) pair tops out around 96.7%, because the same input carries different labels on different rows. The category describes the *case*; the title describes the *charge*. A `THEFT` charge gets filed under *Burglary* often enough that no amount of reading will settle it.

The first run starved its small categories. "In proportion to how often they occur" gives a 25-row category about 14 looks in three epochs. Under 100 training rows, the first run's recall was 17% against the rules' 76%, and 15 named categories never got predicted at all. The fix: draw each category in proportion to the *square root* of its size (that 25-row category now gets about 280 looks), then subtract the tilt back out of the logits so the probabilities still answer for the real class frequencies (Menon et al., 2021). The correction is one number per class, exactly the error the head's bias learned, kept next to the weights where it can be read, checked and removed. The rerun gets 80% on those small categories, leaves none unpredicted, lifts macro-F1 from 0.69 to 0.88, and *gains* three-tenths of a point of accuracy. I picked which run to feature on validation, by a rule written before looking at test: lower validation NLL *and* higher validation rare-class recall. It won both.

The legacy dialect. Half the set-aside `PROMIS Conversion` rows use titles the labeled data never does (`MANUE/DEL 01-15 GR COCAINE/ANLG` is cocaine delivery, spelled the old way). No labels, but confidence still talks. Mean top-class confidence on those rows: title lookup 0.38, statute rules 0.41, TF-IDF 0.72, **the encoder 0.85**. Fixing the small categories made the transformer *more* sure of itself on text it has never seen, which is why the next two sections exist.

🌡️ Calibration: the boring choices did the work



F1: reliability diagram

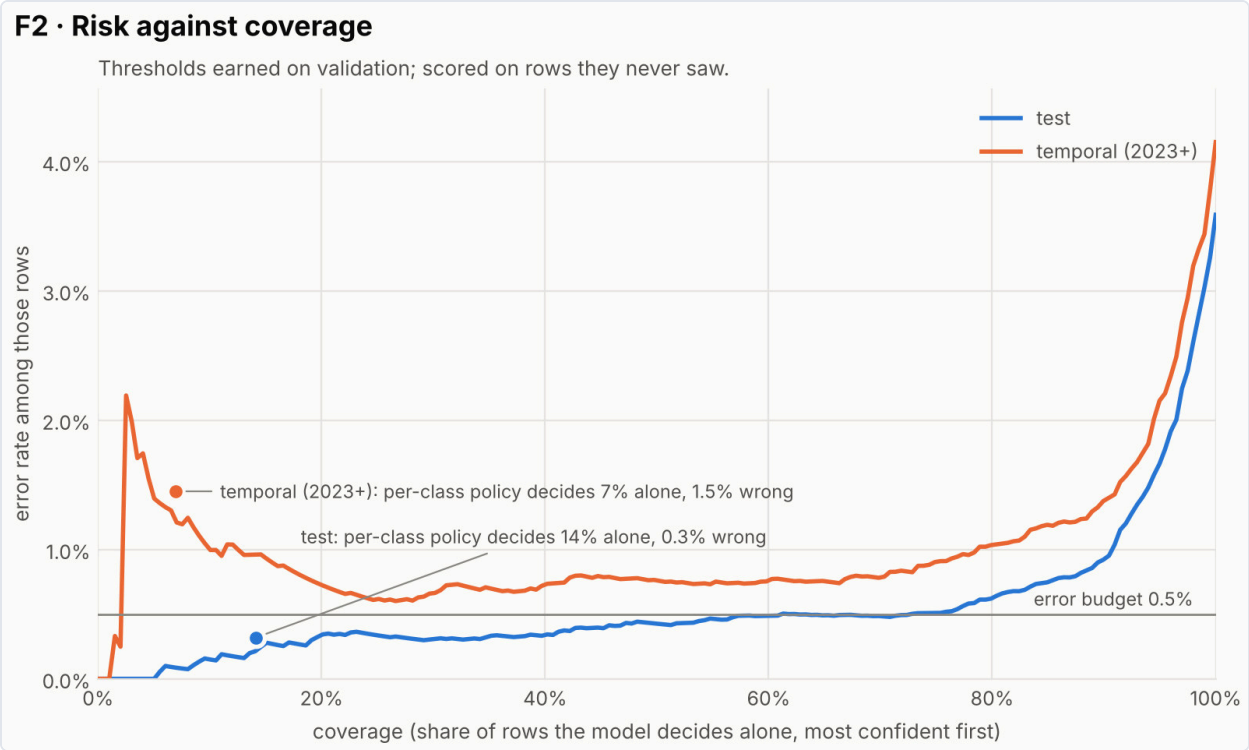
T = 1.00. Temperature scaling found nothing to fix: calibration error is 0.5% on test and 0.9% on 2023+, before and after. Two boring choices had already done the job: picking the epoch by validation NLL, and putting the real prior back with the logit adjustment. The zigzag below 0.6 is small counts: under 3% of rows live there. Two caveats keep this from being a victory lap. Calibrated on familiar data says nothing about a dialect with no labels to check against, and calibrated *on average* isn't calibrated *per class*, which is what the thresholds ask.

🚦 Earned autonomy, at a 0.5% budget

3 of 68 classes earn a threshold:

Category	τ	Test: decided alone	Wrong	2023+: decided alone	Wrong
Narcotics	0.9968	927 of 12,535	3	37 of 6,198	0
Retail Theft	0.9931	1,782 of 1,925	7	811 of 904	7
Aggravated DUI	0.6362	2,002 of 2,013	5	536 of 548	13

Everything else goes to a person. On test, the model decides **14.2%** of rows alone at **0.32%** error. The rerun fixed the small categories and didn't change who earns autonomy: those categories still have far too few validation rows to prove half a percent. Better recall improves what goes to people, not how much the model decides alone. Narcotics, a third of the data and almost always right, earns it on only 927 of its 12,535 test calls: even at the top of the confidence scale its error sits just under the line, and the 95% bound won't round in its favor.

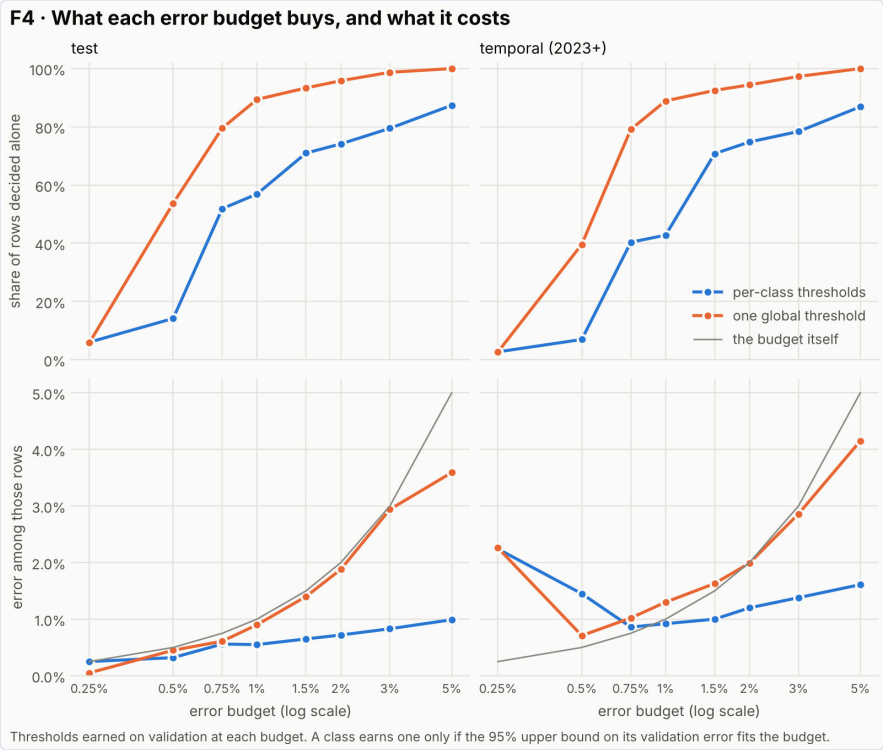


F2: risk–coverage curves

The 2023+ column is the one I'd frame. Thresholds earned on cases from before 2023 decide **6.9%** of later cases alone at **1.45%** error, almost three times the budget, mostly plain DUI charges in cases the office filed under something else. The model didn't change; the world did. Thresholds need an owner and a review date, not just a value.

One global threshold looks better (54% of test rows alone, at 0.45%) until you look inside it: Narcotics runs over budget on test, and four categories run over on 2023+. A global threshold lets the big categories vouch for everybody.

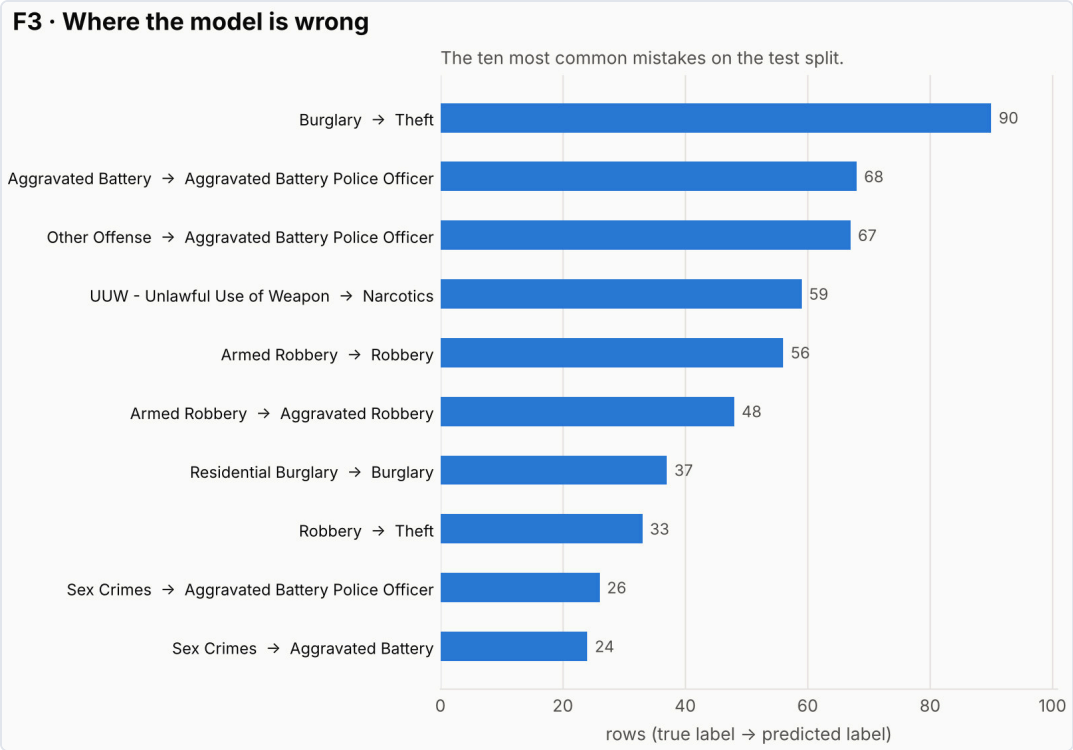
⚖️ The budget is a policy decision



F4: budget sweep

Half a percent is a number I picked, and it sits far below the roughly 3.3% of rows where the labels disagree with themselves. It asks the model to be more consistent than the people who wrote the labels. Relax it and per-class coverage on test climbs: 14% at 0.5%, 57% at 1%, 74% at 2%, 87% at 5%. From 1% up, per-class error stays under the budget on both test sets, while the global threshold runs over it on 2023+ at five of the eight budgets. Choosing the budget is a person's decision. The sweep makes the trade visible; it doesn't make it for you.

🔬 Where it's wrong



F3: top confusions

Two kinds of mistake: neighboring categories (*Armed Robbery* and *Robbery*) and case-level labels on charge-level text (`THEFT` filed as *Burglary*). 91% of the encoder's test errors fall on inputs that already carried more than one label in training, and in 85% of them the model picked the label that input usually gets. Most of its mistakes are the labels disagreeing with themselves.

🌍 The test this data can't run

One county. Every records system spells charges its own way, and the next county's format is the test this dataset can't run; the legacy dialect is a small taste of it. Next: a domain model trained from scratch on this data, and a real cross-jurisdiction test.

✨ Key Takeaways

- **Rules are not weights.** Thirty-six noisy strings belong in a versioned lookup a person signs. `Finding Not Not Guilty` is a typo to fix once, not a pattern to learn.
- **Beat the dictionary before you celebrate.** A statute lookup ties a fine-tuned transformer at 96.4%, under a ceiling of about 96.7% set by labels that describe cases, not charges. The transformer's edge is the small categories, and it only showed up after the rerun.
- **Check how you sample before you blame the model.** Proportional draws starved the small categories (17% recall); square-root draws plus a logit adjustment took them to 80% at no cost in accuracy, and calibration came along for free (T = 1.00).
- **Autonomy is earned per class, and it expires.** Three of 68 categories earned it at 0.5%; on 2023+ cases their error ran nearly three times the budget.
- **Make the budget visible.** A sweep turns a hidden policy choice into a table a person can sign.

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